A BARRAGE OF BARRAGES

Anup Halder, an expert with Vigyan Durbar, a forum that works for the rejuvenation of dying rivers, pegs the Ganga's sediment load at 736 million tonnes a year, 328 MT of which gets deposited upstream of Farakka. When the 2.6-km-long Farakka Barrage was built in 1975 to transfer 40,000 cusecs of water from the Ganga to its distributary, the Bhagirathi Hooghly, its engineers had not planned for such huge quantities of silt, says Dr P.K. Parua, a former general manager of the project.

There is so much silt, in fact, that the 40-km feeder canal, Ahiran Khal, constructed at Dhuliyan in Murshidabad to boost the water flow to wash away the sediment, is proving to be inadequate. Simultaneously, as Halder points out, the saline water coming in from the sea is pushing against the river current, impeding its natural flow and carriage of silt. "The saline water has travelled as far as Triveni in Nadia," he says. It is also destroying the natural vegetation and ecosystem around the river. "A river's natural bank has a variety of grass and trees, with roots going 7-8 metres deep, which helps bind the soil," Halder adds. As this vegetation gets lost, either to salination or to felling, the soil turns vulnerable.

Around Sagar Island, at the mouth of the Gangetic delta, there is 78 times more saline water than freshwater, making a deep flushing of the sediment almost impossible at the Kolkata port. Some four and a half decades later, therefore, the purpose for which Farakka was built—to improve the navigability of the Kolkata port—stands defeated, as silt dredged annually at the port has risen from 6.4 million cubic metres in the pre-Farakka days to 21.9 MCM now, Dr Parua adds.

However, even with the 32 barrages in the Ganga basin across Uttarakhand, West Bengal, Bihar, Jharkhand, Madhya Pradesh, Uttar Pradesh and Haryana facing a similar situation, the Union waterways minister Nitin Gadkari announced that barrages will be built every 100 km along the 1,600-km stretch of the river from Haldia to Allahabad, or 16 new barrages, as part of the Ganga navigation plan. World Bank assistance of $700 million has been sought for various aspects of the plan that the Inland Waterways Authority of India is implementing; the first tranche of $50 million along with technical support has already been promised.

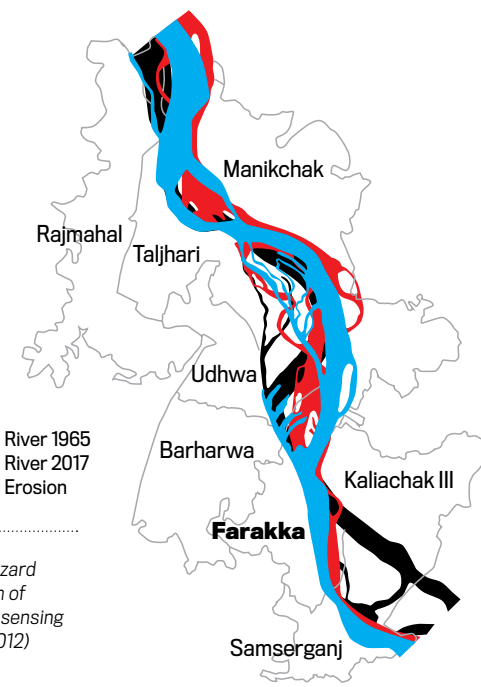


SAJIDA KHATUN, 8

Sitting by the river bank, little Sajida holds her baby sister and looks out to the now calm river, hoping it will return her dolls. Sajida's father and his six brothers from Paschimpara village in Samserganj lost their homes and cultivable lands to the river. When the water had come flooding in the early hours of morning, destroying everything in its path, Sajida had been barely awake. All she remember hearing was some commotion and the next thing she knew, her mother ws dragging her out of the house. Sajida's only regret? That she did not have the chance to save her dolls.

A RIVER'S SHIFTING COURSE

The changed route of the Ganga, upstream and downstream of the Farakka Barrage, and the erosion of its banks



Source: 'River bank erosion hazard study of river Ganga, upstream of Farakka barrage using remote sensing and GIS' by P.K. Thakur et al (2012)